

§ 200. Normal units in H_m .

We now introduce a simplifying notation, by putting

$$r^{5^j} = r_j, \quad (1)$$

and letting β run through a complete system of residues modulo ν . From this it follows that

$$r_{\beta+\nu'} = -r_\beta, \quad (2)$$

and in the form (r, r_j) there are then indeed not all substitutions of the group of the field Ω_m , but the substitutions (r, r_j^{-1}) must still be added; nevertheless (r, r_j) supplies us with all substitutions of the group of H_m .

We shall now denote by $\mathfrak{E}(r)$ a unit of the field H_m . Among these there are also the units of the field H_μ (as imprimitive numbers), and they are characterized by the equation $\mathfrak{E}(r) = \mathfrak{E}(-r)$.

By a **normal unit** of the field H_m we shall now understand a unit which is not equal to ± 1 , but which satisfies the equation

$$\mathfrak{E}(r)\mathfrak{E}(-r) = \pm 1. \quad (3)$$

It is clear that a unit of the field H_μ can certainly not satisfy this condition. But not every unit in H_m which does not belong to the field H_μ will be a normal unit.¹

The units τ_β [§ 199, (5)] do, however, belong to the normal units. Namely,

$$\tau_j = \operatorname{tg} \frac{5^j \pi}{m} = \frac{1 - r_j}{1 + r_j} r_j, \quad (4)$$

and if in this the substitution $(r, -r) = (r_\beta, r_{\beta+\nu'})$ is made, then τ_j goes over into

$$\tau_{\beta+\nu'} = -\frac{1}{\tau_\beta}, \quad (5)$$

which corresponds to relation (3).

A system of ν' normal units

$$\mathfrak{E}_0(r), \mathfrak{E}_1(r), \dots, \mathfrak{E}_{\nu'-1}(r)$$

shall be called an independent system if the determinant

$$\pm \sum \pm \log |\mathfrak{E}_0(r_0)| \log |\mathfrak{E}_1(r_1)| \cdots \log |\mathfrak{E}_{\nu'-1}(r_{\nu'-1})| = L(\mathfrak{E}_0, \mathfrak{E}_1, \dots, \mathfrak{E}_{\nu'-1}) \quad (6)$$

has a value different from zero.

The absolute value of this determinant, L , we here also call the **regulator** of the system $\mathfrak{E}_0, \mathfrak{E}_1, \dots, \mathfrak{E}_{\nu'-1}$.

The units $\tau_0, \tau_1, \dots, \tau_{\nu'-1}$ form an independent system of normal units; for their regulator

$$L(\tau_0, \tau_1, \dots, \tau_{\nu'-1})$$

is just the determinant T defined in the preceding paragraph [§ 199, (14), (15)]. And that this cannot be zero follows immediately from the fact that it occurs as a factor in the expression for the class number. But the class number, since at least one class, namely the principal class, certainly always exists, is different from zero; hence T too cannot be equal to zero. We therefore have the theorem:

¹In the above-mentioned work (Acta mathematica, vol. 8), the author designated the normal units as “primitive units”. This name is rejected here, because it could give occasion to the error that every unit of the field H_m which does not at the same time belong to the field H_μ is to be counted among them.

1. The numbers $\tau_0, \tau_1, \dots, \tau_{\nu'-1}$ are an independent system of normal units of the field H_m .

If $\mathfrak{E}(r)$ is a normal unit, then, putting

$$l_\beta = \log |\mathfrak{E}(r_\beta)|, \quad (7)$$

one has, because of (3),

$$l_{\beta+\nu'} = -l_\beta. \quad (8)$$

If now $\mathfrak{E}_0, \mathfrak{E}_1, \dots, \mathfrak{E}_{\nu'-1}$ is any independent system of normal units, we put

$$l_{i\beta} = \log |\mathfrak{E}_i(r_\beta)|, \quad l_{i,\beta+\nu'} = -l_{i\beta}, \quad (9)$$

so that the regulator of the system of the $\mathfrak{E}_i$ receives the expression

$$\pm \sum \pm l_{00} l_{11} \cdots l_{\nu'-1, \nu'-1} = L(\mathfrak{E}_0, \mathfrak{E}_1, \dots, \mathfrak{E}_{\nu'-1}) \quad (10)$$

and is a number different from zero, which we denote by $T_{\mathfrak{E}}$.

If $\mathfrak{E}(r)$ denotes any other normal unit, then we can determine the unknowns $\xi_0, \xi_1, \dots, \xi_{\nu'-1}$ from the ν' linear equations which we obtain when, in

$$l_\beta = \xi_0 l_{0\beta} + \xi_1 l_{1\beta} + \cdots + \xi_{\nu'-1} l_{\nu'-1, \beta}, \quad (11)$$

we let the index β run from 0 to $\nu' - 1$; and because of (8) and (9), this equation is also correct for the remaining values of β , so that from it one may derive the formula, valid for all r :

$$\mathfrak{E}(r) = \pm \mathfrak{E}_0(r)^{\xi_0} \mathfrak{E}_1(r)^{\xi_1} \cdots \mathfrak{E}_{\nu'-1}(r)^{\xi_{\nu'-1}}. \quad (12)$$

Conversely, if $m_0, m_1, \dots, m_{\nu'-1}$ denote arbitrary rational integers, then all numbers contained in the form

$$\pm \mathfrak{E}_0(r)^{m_0} \mathfrak{E}_1(r)^{m_1} \cdots \mathfrak{E}_{\nu'-1}(r)^{m_{\nu'-1}}$$

are normal units of the field H_m . Consequently the considerations which we carried out in detail in § 186 may be applied to the system of equations (10), and this gives:

2. The exponents $\xi_0, \xi_1, \dots, \xi_{\nu'-1}$ determined from (10) or (11) are rational numbers.

From this one may infer, as in § 187, that there are systems of ν' independent normal units whose regulator T_0 is as small as possible, and such systems are called **fundamental systems of normal units**.

If $\mathfrak{E}_0, \mathfrak{E}_1, \dots, \mathfrak{E}_{\nu'-1}$ is such a fundamental system, then the exponents $\xi_0, \xi_1, \dots, \xi_{\nu'-1}$ in (10) and (11) are obtained as **rational integers**. This too can be proved in the same way as in § 187.

If, under this assumption concerning the $\mathfrak{E}_i$, we set up the equations (11) for any other system of independent units $\mathfrak{E}_0^{(1)}, \mathfrak{E}_1^{(1)}, \dots, \mathfrak{E}_{\nu'-1}^{(1)}$, then we obtain expressions of the form

$$L_{i,k}^{(1)} = \xi_{i,0} L_{0,k} + \xi_{i,1} L_{1,k} + \cdots + \xi_{i,\nu'-1} L_{\nu'-1,k}.$$

Thus, if T_1 is the regulator of the system of the $\mathfrak{E}_i^{(1)}$, then, by the multiplication theorem for determinants, if C denotes the absolute value of the determinant of the integral exponents $\xi_{i,k}$, hence a positive rational integer, there results

$$T_1 = CT_0. \quad (13)$$

It cannot be proved, and for the higher values of m probably also is not true, that $\tau_0, \tau_1, \dots, \tau_{\nu'-1}$ form a fundamental system.²

For our purpose, however, it is important that these units behave, with respect to the denominator 2 in the exponents, in a certain sense like a fundamental system; this is expressed more precisely by the following theorem:

2. If a normal unit of the field H_m is of the form

$$\mathfrak{E}(r) = \tau_0^{e_0} \tau_1^{e_1} \cdots \tau_{\nu'-1}^{e_{\nu'-1}},$$

where the exponents $e_0, e_1, \dots, e_{\nu'-1}$ are rational integers, and if it has one and the same sign with all its conjugate values, then all $e_0, e_1, \dots, e_{\nu'-1}$ must be even numbers, and $\mathfrak{E}(r)$ is the square of a normal unit.

We can again conduct the proof by complete induction; but first we shall transform the expression for τ_j somewhat.

The conjugate values of the unit $\mathfrak{E}(r)$ are obtained in the form

$$\mathfrak{E}(r_\beta) = \tau_\beta^{e_0} \tau_{\beta+1}^{e_1} \cdots \tau_{\beta+\nu'-1}^{e_{\nu'-1}},$$

and these expressions are therefore supposed to have the same sign for all values of β .

By § 199, (5),

$$\tau_j = \operatorname{tg} \frac{5^j \pi}{m} = -\frac{1}{2} \frac{\sin \left(\frac{2\pi}{m} 5^j \right)}{\left[\cos \left(\frac{5^j \pi}{m} \right) \right]^2}.$$

Here, for abbreviation, we put

$$\sigma_j = \sin \left(\frac{2\pi}{m} 5^j \right), \quad (14)$$

so that σ_j always has the same sign as τ_j .

From this we form the product

$$S_\beta = \sigma_\beta^{e_0} \sigma_{\beta+1}^{e_1} \cdots \sigma_{\beta+\nu'-1}^{e_{\nu'-1}}, \quad (15)$$

and our assumption is therefore that S_β has the same sign for all values of β .

It is to be shown that the rational integers $e_0, e_1, \dots, e_{\nu'-1}$ must all be even.

The theorem is correct for $m = 8$; for in this case

$$\sigma_0 = \sin \frac{2\pi}{8}, \quad \sigma_1 = \sin \frac{10\pi}{8} = -\sin \frac{6\pi}{8},$$

therefore σ_0 is positive and σ_1 negative. Here $S_\beta = \sigma_\beta^{e_0}$, and S_0 and S_1 can have the same sign only when e_0 is even.

After this we apply complete induction. For simplification we put $\nu' = 2\nu''$ and recall the notation

$$m = 2^\kappa, \quad \mu = \frac{1}{2}m, \quad \nu = \frac{1}{2}\mu, \quad \nu' = \frac{1}{2}\nu, \quad \nu'' = \frac{1}{2}\nu', \quad (16)$$

²Here, as in all questions concerning units, it is very difficult to arrive at definite numerical results. According to the results contained in the rich work of Reuschle, *Tafeln complexer Primzahlen, welche aus Wurzeln der Einheiten gebildet sind* (Berlin 1875), for $m = 16$ the τ still form a fundamental system. For here the basic number of the field H_m is equal to 2^{11} , the degree of the field is equal to 4, and hence, by the note to § 156, in every ideal class there is an ideal whose norm is less than 6. But according to Reuschle's tables every natural prime number below 100 can be decomposed in the field H_m into actually existing prime factors. Consequently in all numbers below 6 only principal ideals certainly occur, and consequently there is here only the one class, the principal class. Since, as we have seen, the first class-number factor too is equal to 1, the cyclotomic field Ω_{16} is of one class. The same laws of primes hold in it as in the field of rational numbers. This is doubtful for the field Ω_{32} and certainly no longer applies in the field Ω_{64} , in which the class number is a multiple of 17.

and assume that our theorem has already been proved when μ takes the place of m .

Then

$$\begin{aligned} 5^\nu &\equiv 1 \pmod{m}, & 5^{\nu'} &\equiv 1 + \mu \pmod{m}, & 5^{\nu''} &\equiv 1 + \nu \pmod{\mu}, \\ 5^{\nu''} &\equiv 1 + \nu + \mu \pmod{m}, \end{aligned}$$

and from this it follows that

$$\begin{aligned} \sigma_{\beta+\nu} &= -\sigma_\beta, \\ \sigma_{\beta+\nu''} &= -\cos\left(\frac{2\pi}{m}5^\beta\right), \\ \sigma_\beta\sigma_{\beta+\nu''} &= -\frac{1}{2}\sin\left(\frac{2\pi}{\mu}5^\beta\right) = -\frac{1}{2}\sigma'_\beta, \end{aligned} \tag{17}$$

$$\sigma'_{\beta+\nu''} = -\sigma'_\beta, \tag{18}$$

where the σ'_β arise from the σ_β by the transition from κ to $\kappa - 1$. We now form, according to (17), (18), the product $S_\beta S_{\beta+\nu''}$, and obtain, if for abbreviation we also put

$$\begin{aligned} e' &= e_0 + e_1 + \dots + e_{\nu'-1}, & e'' &= e_0 + e_1 + \dots + e_{\nu''-1}, \\ S_\beta S_{\beta+\nu''} &= (-1)^{\nu''} \left(\frac{1}{2}\right)^{e'} (\sigma'_\beta)^{e_0+e_{\nu''}} (\sigma'_{\beta+1})^{e_1+e_{\nu''+1}} \dots (\sigma'_{\beta+\nu''-1})^{e_{\nu''-1}+e_{\nu'-1}}. \end{aligned}$$

This, however, is a product of the same form as S_β , with μ in place of m , and it likewise has the same sign with all its conjugates. Therefore by the assumption made,

$$e_0 \equiv e_{\nu''}, \quad e_1 \equiv e_{\nu''+1}, \dots, e_{\nu''-1} \equiv e_{\nu'-1} \pmod{2}. \tag{19}$$

Putting, then,

$$\begin{aligned} S'_\beta &= (\sigma'_\beta)^{e_0} (\sigma'_{\beta+1})^{e_1} \dots (\sigma'_{\beta+\nu''-1})^{e_{\nu''-1}}, \\ R_\beta &= \sigma_{\beta+\nu''}^{\frac{1}{2}(e_0-e_{\nu''})} \sigma_{\beta+\nu''+1}^{\frac{1}{2}(e_1-e_{\nu''+1})} \dots \sigma_{\beta+\nu'-1}^{\frac{1}{2}(e_{\nu''-1}-e_{\nu'-1})}, \end{aligned}$$

it follows from (15) and (17) that

$$S_\beta R_\beta = \left(-\frac{1}{2}\right)^{e''} S'_\beta,$$

and hence S'_β , just as S_β , has the same sign with all its conjugate values. But S'_β arises from S_β by putting μ in place of m , and therefore by our assumption

$$e_0 \equiv 0, \quad e_1 \equiv 0, \dots, e_{\nu''-1} \equiv 0 \pmod{2},$$

and, with the aid of (19),

$$e_{\nu''} \equiv 0, \quad e_{\nu''+1} \equiv 0, \dots, e_{\nu'-1} \equiv 0 \pmod{2},$$

which completes the proof of Theorem 2.

If now in formula (12) for $\mathfrak{E}_0, \mathfrak{E}_1, \dots, \mathfrak{E}_{\nu'-1}$ we choose the system $\tau_0, \tau_1, \dots, \tau_{\nu'-1}$, and put the exponents ξ into the form

$$\xi_0 = \frac{e_0}{e}, \quad \xi_1 = \frac{e_1}{e}, \dots, \quad \xi_{\nu'-1} = \frac{e_{\nu'-1}}{e},$$

where $e, e_0, e_1, \dots, e_{\nu'-1}$ are rational integers without a common divisor, then we obtain

$$\mathfrak{E}(r)^e = \pm \tau_0^{e_0} \tau_1^{e_1} \dots \tau_{\nu'-1}^{e_{\nu'-1}}.$$

It follows from our theorem that e must be odd; for if it were even, then $\mathfrak{E}(r)^e$ would be the square of a real quantity, and consequently $\pm \mathfrak{E}(r)^e$ would have the same sign with its conjugates. But then, according to our theorem, all $e_0, e_1, \dots, e_{\nu'-1}$ would have to be divisible by 2, contrary to the assumption that the exponents $e, e_0, \dots, e_{\nu'-1}$ should have no common divisor. We therefore have the theorem:

- 3. If $\mathfrak{E}(r)$ is any normal unit of the field H_m , then the odd number e and the rational integers $e_0, e_1, \dots, e_{\nu'-1}$ can be so determined that**

$$\mathfrak{E}(r_\beta)^e = \pm \tau_\beta^{e_0} \tau_{\beta+1}^{e_1} \cdots \tau_{\beta+\nu'-1}^{e_{\nu'-1}} \quad (20)$$

is obtained.

This result leads to some important consequences:

- 4. If T is the regulator of the system of the τ_j , and T_0 the regulator of a fundamental system of normal units in H_m , i.e. the minimal value which the regulator of any system of independent normal units can assume, then**

$$T = CT_0, \quad C \equiv 1 \pmod{2}, \quad (21)$$

where C denotes an odd natural number.

For the fact that $T : T_0 = C$ is an integer follows from formula (13).

If, however, we apply Theorem 3 to all elements of the fundamental system $\mathfrak{E}_i(r)$, then a system of integers $e_{i,k}$ and an odd number e result such that

$$\mathfrak{E}_i(r)^e = \pm \tau_0^{e_{i,0}} \tau_1^{e_{i,1}} \cdots \tau_{\nu'-1}^{e_{i,\nu'-1}}.$$

Taking logarithms of these equations, and then forming the regulator T_0 , we obtain

$$e^{\nu'} T_0 = \pm \sum \pm e_{0,0} e_{1,1} \cdots e_{\nu'-1, \nu'-1} T,$$

hence

$$e^{\nu'} = \pm C \sum \pm e_{0,0} e_{1,1} \cdots e_{\nu'-1, \nu'-1},$$

and since in this equation the left-hand side is odd, C cannot be even, as was to be proved.

If all conjugate values of the normal unit $\mathfrak{E}(r_\beta)$ have one and the same sign, then the same is also true of $\mathfrak{E}(r_\beta)^e$, and consequently in this case the exponents $e_0, e_1, \dots, e_{\nu'-1}$ in formula (20) must be even numbers by 2. If one then writes formula (20) in the form

$$\mathfrak{E}(r) = \pm \tau_0^{e_0} \tau_1^{e_1} \cdots \tau_{\nu'-1}^{e_{\nu'-1}} \mathfrak{E}(r)^{1-e},$$

one obtains from it the theorem:

- 5. A normal unit of the field H_m which has the same sign with its conjugates is, apart from its sign, the square of a normal unit.**

§ 201. Fundamental system of units of the field H_m .

It remains to determine the denominator D in the formula for B [§ 197, (6)], which was defined by the equation

$$E = E' D,$$

where E and E' denoted the regulators of the fields H_m, H_μ .

Here, therefore, the matter is no longer normal units, but units of the field H_m in general.

We assume a fundamental system of units in the field H_m :

$$\varepsilon_1(r), \varepsilon_2(r), \dots, \varepsilon_{\nu-1}(r), \quad (1)$$

and let the conjugate logarithms of this system be

$$l_{1,k}, l_{2,k}, \dots, l_{\nu-1,k}, \quad k = 0, 1, 2, \dots, \nu - 1. \quad (2)$$

Alongside this we consider a fundamental system in the field H_μ ,

$$\varepsilon'_1, \varepsilon'_2, \dots, \varepsilon'_{\nu'-1}, \quad (3)$$

with the conjugate logarithms

$$l'_{1,k}, l'_{2,k}, \dots, l'_{\nu'-1,k}, \quad k = 0, 1, 2, \dots, \nu' - 1. \quad (4)$$

In forming the regulators, one of the conjugate values, say $k = 0$, is omitted (cf. § 186), and therefore we obtain

$$E = \pm \sum \pm l_{1,1} l_{2,2} \cdots l_{\nu-1,\nu-1}, \quad E' = \pm \sum \pm l'_{1,1} l'_{2,2} \cdots l'_{\nu'-1,\nu'-1}. \quad (5)$$

Now we add to the system (3) a fundamental system of normal units of the field H_m ,

$$\mathfrak{E}_0, \mathfrak{E}_1, \dots, \mathfrak{E}_{\nu'-1}, \quad (6)$$

and assert that the system

$$\varepsilon'_1, \varepsilon'_2, \dots, \varepsilon'_{\nu'-1}, \mathfrak{E}_0, \mathfrak{E}_1, \dots, \mathfrak{E}_{\nu'-1} \quad (7)$$

is an independent system of units of the field H_m (even if not a fundamental system).

If we denote by

$$L_{0,k}, L_{1,k}, \dots, L_{\nu'-1,k}, \quad k = 0, 1, \dots, \nu' - 1 \quad (8)$$

the conjugate logarithms of the system (6), then by the preceding paragraph

$$\pm \sum \pm L_{0,0} L_{1,1} \cdots L_{\nu'-1,\nu'-1} = T_0 \quad (9)$$

is the regulator of this system; and in view of the definition of normal units [§ 200, (9)] we have

$$L_{i,k+\nu'} = -L_{i,k}, \quad (10)$$

and since the ε'_i as numbers of the field H_μ remain unchanged by the substitution $(r, -r)$,

$$l'_{i,k+\nu'} = l'_{i,k}. \quad (11)$$

Accordingly the regulator R of the system (7) is obtained from the determinant

$$\begin{vmatrix} l'_{1,1} & l'_{2,1} & \cdots & l'_{\nu'-1,1} \\ \vdots & \vdots & & \vdots \\ l'_{1,\nu'-1} & l'_{2,\nu'-1} & \cdots & l'_{\nu'-1,\nu'-1} \\ l'_{1,0} & l'_{2,0} & \cdots & l'_{\nu'-1,0} \\ l'_{1,1} & l'_{2,1} & \cdots & l'_{\nu'-1,1} \\ \vdots & \vdots & & \vdots \\ l'_{1,\nu'-1} & l'_{2,\nu'-1} & \cdots & l'_{\nu'-1,\nu'-1} \end{vmatrix} \begin{vmatrix} L_{0,1} & L_{1,1} & \cdots & L_{\nu'-1,1} \\ \vdots & \vdots & & \vdots \\ L_{0,\nu'-1} & L_{1,\nu'-1} & \cdots & L_{\nu'-1,\nu'-1} \\ -L_{0,0} & -L_{1,0} & \cdots & -L_{\nu'-1,0} \\ -L_{0,1} & -L_{1,1} & \cdots & -L_{\nu'-1,1} \\ \vdots & \vdots & & \vdots \\ -L_{0,\nu'-1} & -L_{1,\nu'-1} & \cdots & -L_{\nu'-1,\nu'-1} \end{vmatrix}.$$

This determinant has $\nu - 1$ rows. We add the first to the $(\nu' + 1)$ -st, the second to the $(\nu' + 2)$ -nd, and so on, the $(\nu' - 1)$ -st to the last. In the last $(\nu' - 1)$ rows the L thereby cancel, and the l' acquire the factor 2 [by (10), (11)]; and by a determinant theorem (vol. I, § 23, V.) this gives

$$R = 2^{\nu'-1} E' T_0. \quad (12)$$

Since, accordingly, R does not vanish, and hence in (7) there is a system of $\nu - 1$ independent units of the field H_m , we can [by § 186, (5)] determine the rational (integral or fractional) numbers $m_{i,k}, M_{i,k}$ from the equations

$$2l_{i,k} = m_{1,i}l'_{1,k} + \cdots + m_{\nu'-1,i}l'_{\nu'-1,k} + M_{0,i}L_{0,k} + M_{1,i}L_{1,k} + \cdots + M_{\nu'-1,i}L_{\nu'-1,k}. \quad (13)$$

But it is easily shown, by means of the relations (10), (11), that the numbers $m_{i,k}, M_{i,k}$ here must be **integral**. For

$$l_{i,k} + l_{i,k+\nu'} = \log |\varepsilon_i(r_k)\varepsilon_i(-r_k)| = m_{1,i}l'_{1,k} + \cdots + m_{\nu'-1,i}l'_{\nu'-1,k},$$

$$l_{i,k} - l_{i,k+\nu'} = \log |\varepsilon_i(r_k)\varepsilon_i(-r_k)^{-1}| = M_{0,i}L_{0,k} + M_{1,i}L_{1,k} + \cdots + M_{\nu'-1,i}L_{\nu'-1,k}.$$

But the units $\varepsilon_i(r)\varepsilon_i(-r)$ belong to the field H_μ , and since by assumption the ε'_i form a fundamental system of this field, the $m_{i,k}$ must be rational integers by § 187.

The units $\varepsilon_i(r)\varepsilon_i(-r)^{-1}$ are normal units of the field H_m , and since the $\mathfrak{E}_i$ are supposed to be a fundamental system of normal units, the $M_{i,k}$ are integers by § 200.

For better survey we put the equations (13), once again without the index k which distinguishes the conjugate values from one another, here:

$$2l_i = m_{1,i}l'_1 + m_{2,i}l'_2 + \cdots + m_{\nu'-1,i}l'_{\nu'-1} + M_{0,i}L_0 + M_{1,i}L_1 + \cdots + M_{\nu'-1,i}L_{\nu'-1}. \quad (14)$$

If we now form the regulator E according to (13), and denote by M the absolute value of the determinant of the $\nu - 1$ rows of the numbers $m_{k,i}, M_{k,i}$,

$$M = \pm \sum \pm m_{1,1} \cdots m_{\nu'-1,\nu'-1} M_{0,0} M_{1,1} \cdots M_{\nu'-1,\nu'-1},$$

then there follows

$$2^{\nu-1}E = MR, \quad (15)$$

and from this, in view of (12),

$$E = 2^{-\nu'} M E' T_0. \quad (16)$$

It can now further be shown that M is a power of 2 and is divisible by $2^{\nu'}$.

For since the system (1) was assumed to be a fundamental system in H_m , there are rational integers $n_{k,i}, N_{k,i}$ satisfying the equations

$$\begin{aligned} l'_i &= n_{1,i}l_1 + n_{2,i}l_2 + \cdots + n_{\nu-1,i}l_{\nu-1}, & i &= 1, 2, \dots, \nu' - 1, \\ L_i &= N_{1,i}l_1 + N_{2,i}l_2 + \cdots + N_{\nu-1,i}l_{\nu-1}, & i &= 0, 1, 2, \dots, \nu' - 1. \end{aligned} \quad (17)$$

If from these we form the determinant R , and denote by N the absolute value of the determinant of the $n_{k,i}, N_{k,i}$, then

$$R = NE,$$

and hence by (15)

$$MN = 2^{\nu-1};$$

from this it follows that each of the two integers M, N must be a power of 2.

To decide by which power of 2 the determinant M is divisible, we first determine a system of rational integers a_i without common divisor which satisfies the linear equations

$$\sum_{i=1}^{\nu-1} a_i m_{k,i} = 0 \quad (k = 1, 2, \dots, \nu' - 1). \quad (18)$$

If we then put

$$\sum_{i=1}^{\nu-1} a_i M_{k,i} = \xi_k,$$

we obtain from the equations (13)

$$2 \sum_{i=1}^{\nu-1} a_i l_{i,k} = \xi_0 L_{0,k} + \xi_1 L_{1,k} + \cdots + \xi_{\nu'-1} L_{\nu'-1,k}.$$

From this one obtains by (10)

$$\sum_{i=1}^{\nu-1} a_i l_{i,k+\nu'} = - \sum_{i=1}^{\nu-1} a_i l_{i,k},$$

and this means that

$$\varepsilon_1^{a_1} \varepsilon_2^{a_2} \cdots \varepsilon_{\nu-1}^{a_{\nu-1}}$$

is a normal unit of the field H_m . It then follows further, since $\mathfrak{E}_0, \mathfrak{E}_1, \dots, \mathfrak{E}_{\nu'-1}$ is a fundamental system of normal units, that the numbers $\xi_0, \xi_1, \dots, \xi_{\nu'-1}$ must be even. Hence:

1. The system of equations (18) implies the congruences

$$\sum_{i=1}^{\nu-1} a_i M_{k,i} \equiv 0 \pmod{2} \quad (k = 0, 1, \dots, \nu' - 1). \quad (19)$$

But in the equations (18), whose number is only $\nu' - 1$, there are $\nu - 1$ unknowns a_i . Therefore there are several independent systems of solutions, which we may state more precisely as follows:

2. There are ν' integral solutions of the system (18):

$$\begin{array}{cccc} a_{1,1} & a_{2,1} & \cdots & a_{\nu-1,1} \\ a_{1,2} & a_{2,2} & \cdots & a_{\nu-1,2} \\ \cdots & \cdots & & \cdots \\ a_{1,\nu'} & a_{2,\nu'} & \cdots & a_{\nu-1,\nu'} \end{array} \quad (20)$$

such that from the matrix (20) a determinant of ν' rows can be formed which is an odd number.

For first there is a solution of (18),

$$a_{1,1}, a_{2,1}, \dots, a_{\nu-1,1},$$

in which an odd number occurs. Suppose, as is permitted, since the ordering of the indices is immaterial here, that this odd number is $a_{1,1}$; then we determine a second solution

$$0, a_{2,2}, a_{3,2}, \dots, a_{\nu-1,2},$$

in which again an odd number, say $a_{2,2}$, occurs. Then we determine a third solution

$$0, 0, a_{3,3}, \dots,$$

and continue setting equal to zero as long as the number of remaining unknowns is still greater than the number of equations.

In the last system, therefore, $\nu' - 1$ zeros will occur, leaving ν' unknowns.

We can take these rows of numbers $a_{i,k}$ for the system (20), for then the determinant

$$\pm \sum \pm a_{1,1} a_{2,2} \cdots a_{\nu',\nu'} \quad (21)$$

reduces to the diagonal term $a_{1,1} a_{2,2} \cdots a_{\nu',\nu'}$, all whose factors are odd.

If now the requirement of Theorem 2 is fulfilled, the matrix (20) can be completed, by adding $\nu' - 1$ rows, to a determinant of $\nu - 1$ rows which likewise has an odd integral value:

$$\alpha = \pm \sum \pm a_{1,1} a_{2,2} \cdots a_{\nu',\nu'} a_{\nu'+1,\nu'+1} \cdots a_{\nu-1,\nu-1}. \quad (22)$$

It is only necessary, when the determinant (21) is assumed to be odd, to put zeros throughout the rows that have been added, except for the elements in the diagonal row, $a_{\nu'+1,\nu'+1}, \dots, a_{\nu-1,\nu-1}$, which may be taken equal to 1. Then α receives the same value as the determinant (21).

But if now, by the multiplication theorem for determinants, we form the product αM , then there results a determinant in which the $\nu - 1$ sums

$$\sum_{i=1}^{\nu-1} a_i m_{k,i}, \quad \sum_{i=1}^{\nu-1} a_i M_{k,i},$$

$$k = 1, 2, \dots, \nu' - 1, \quad k = 0, 1, 2, \dots, \nu' - 1$$

form a row for each value $s = 1, 2, \dots, \nu'$; and in it, therefore, ν' rows occur whose elements are all divisible by 2.

Consequently αM , and therefore also M , is divisible by $2^{\nu'}$, and if we put

$$M = 2^{\nu'+\theta}, \quad N = 2^{\nu'-\theta-1}, \quad (23)$$

then θ is a non-negative rational integer.

Putting this into (16) gives

$$E = 2^\theta E' T_0.$$

But in § 197, (6), one had put

$$E = E' D,$$

and hence it follows that

$$D = 2^\theta T_0.$$

From this, by § 199, (15) and § 200, (21), one obtains for the class-number factor B

$$B = 2^{-\theta} \frac{T}{T_0} = 2^{-\theta} C, \quad (24)$$

where C is an odd integer.

Accordingly, by § 197, (4), for the second factor of the class number h_1 , which, as we have seen, is an integer, one obtains

$$h_1 = 2^{-\theta} h'_1 C. \quad (25)$$

If we now assume as proved that h'_1 is odd, then, since C also is odd, it follows that $\theta = 0$ and h_1 is odd; and both are thus proved by complete induction.

The derived results are thereby simplified as follows:

3. One has

$$M = 2^{\nu'}, \quad D = T_0, \quad B = \frac{T}{T_0}; \quad (26)$$

the class-number factor B is equal to the ratio of the regulator T of the units τ to the regulator T_0 of a fundamental system of normal units, and is an odd integer.

If we denote by B_κ the number B belonging to $m = 2^\kappa$, then the class number h_1 , as follows by complete induction, is

$$h_1 = B_4 B_5 \cdots B_\kappa, \quad (27)$$

and hence is also an odd number. Thus, with regard to § 198, 1., the main theorem is proved:

A. The class number in the field Ω_m , when m is a power of 2, is odd.

§ 202. Positive units.

The results found last show that the system of independent units § 201, (7),

$$\varepsilon'_1, \varepsilon'_2, \dots, \varepsilon'_{\nu'-1}, \mathfrak{E}_0, \mathfrak{E}_1, \dots, \mathfrak{E}_{\nu'-1},$$

in the field H_m is not a fundamental system. For otherwise in the formulae § 201, (13), all the numbers $m_{k,i}, M_{k,i}$ would have to be divisible by 2, and the determinant M , whose value we found to be $2^{\nu'}$, would be divisible by $2^{\nu'-1}$.

In place of the system of equations (18), § 201, we now consider the following congruences:

$$\begin{aligned} \sum_{i=1}^{\nu-1} b_i m_{1,i} &\equiv 1 \pmod{2}, \\ \sum_{i=1}^{\nu-1} b_i m_{2,i} &\equiv 0 \pmod{2}, \\ &\vdots \\ \sum_{i=1}^{\nu-1} b_i m_{\nu'-1,i} &\equiv 0 \pmod{2}, \end{aligned} \tag{1}$$

and prove that it is possible to satisfy them by integral values of the b_i .

If this were not possible, then from the $\nu' - 2$ congruences

$$\sum_{i=1}^{\nu-1} b_i m_{k,i} \equiv 0 \pmod{2} \quad k = 2, 3, \dots, \nu' - 1$$

the last congruence

$$\sum_{i=1}^{\nu-1} b_i m_{1,i} \equiv 0 \pmod{2}$$

would have to follow; and then, as in the preceding paragraph, it would further follow that

$$\sum_{i=1}^{\nu-1} b_i M_{k,i} \equiv 0 \pmod{2} \quad k = 0, 1, \dots, \nu' - 1.$$

Then in place of the matrix § 201, (20), one could put a matrix of the $b_{k,i}$ with $\nu' + 1$ rows, and on the path taken in the preceding paragraph it would follow that M would have to be divisible by $2^{\nu'+1}$, which is not the case.

But if we apply the system of multipliers b_i which satisfies the conditions (1) to the equations (14) of the preceding paragraph, multiplying by b_i and then summing, there results, when we omit multiples of 2 and express this also here, where irrational numbers, namely logarithms, occur, by the congruence sign:

$$l'_1 \equiv L_0 \sum M_{0,i} b_i + L_1 \sum M_{1,i} b_i + \dots + L_{\nu'-1} \sum M_{\nu'-1,i} b_i \pmod{2}. \tag{2}$$

If we wish to transform the congruence into an equation, then a sum of quantities

$$l_1, l_2, \dots, l_{\nu-1}, \quad l'_1, l'_2, \dots, l'_{\nu'-1}$$

is to be added, with even integers as coefficients.

The same consideration may also be applied to $l'_2, l'_3, \dots, l'_{\nu'-1}$; and from this, when one passes back from the logarithms to the numbers, formula (2) may be expressed in words as follows:

1. Every unit of the field H_μ is representable as the product of a normal unit of the field H_m and the square of a unit in H_m .

If we now apply Theorem 2, § 200 to this, then it follows that every unit $\mathfrak{E}(r)$ of the field H_μ which has the same sign with all its conjugates must, apart from sign, be the square of a unit in H_m . But if

$$\pm \mathfrak{E}(r) = \Delta(r)^2,$$

where $\mathfrak{E}(r)$ is a unit in H_μ and $\Delta(r)$ a unit in H_m , then $\Delta(r)$ too must be contained in H_μ .

For if we put

$$\Delta(r) = \Delta_1(r^2) + r\Delta_2(r^2),$$

then its square can be contained in H_μ , and so remain unchanged under the change of sign of r , only if either Δ_1 or $\Delta_2 = 0$. But it cannot be that $\Delta(r) = r\Delta_2(r^2)$, for otherwise $\Delta_2(r^2)$ would be a unit in Ω_μ and would have to be, by § 178, 1., the product of a real unit and a root of unity $r^{2\lambda}$; and since $\Delta(r)$ too is real, $r^{2\lambda+1}$ would have to be real, which is impossible. Hence $\Delta(r)$ is contained in H_μ .

Since now μ , as well as m , can be any power of 2, the second main theorem follows from this:

B. A unit of the field H_m which is positive with all its conjugates is the square of a unit of the same field.

On the two theorems A. and B., however, in § 184 we made the proof of the main theorem on Abelian number fields (§ 179, I.) depend.